



DEPARTMENT OF CHEMICAL ENGINEERING NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Bioresource Technology (CET-465)

Home Assignment plus Question Bank-4

S. No.	Questions	COs
1.	What is single cell proteins (SCP) and why is there need for it? Discuss its advantages and disadvantages.	CO1
2.	What are various suitable substrates which can be used for production of SCP? Discuss the sources of microorganisms and the selection criteria for SCP production.	CO1
3.	Write a brief note on the nutritional aspects of SCP.	CO2
4.	Explain different factors which effect the production of SCP.	CO3
5.	Discuss single cell protein production processes along with a suitable flow diagram.	CO3
6.	What are the various steps involved and technologies applied worldwide for production of SCP? Discuss the Pruteen Technology used.	CO3
7.	Briefly describe the uses of SCP.	CO2
8.	How can the production of SCP be optimized? Discuss the advantages and drawbacks of SCP technologies.	CO3
9.	Write a brief note on antibiotics and discuss their top ten classes.	CO2
10.	Explain the different schemes for classification of antibiotics with suitable examples.	CO2
11.	Explain the chemical structure of antibiotics with the help of cephalosporin and penicillin.	CO2
12.	What is penicillin? Discuss the history behind the discovery and production.	CO1
13.	Explain the general structure of penicillin and discuss the significance of R and M groups in the side chains.	CO2
14.	What is an ideal penicillin? Discuss its classification based on the mode of production. Also describe the side effects.	CO2
15.	Give at least ten examples of penicillins and discuss their side effects.	CO2
16.	What is antibiotic resistance and what is the opinion of WHO on it?	CO2
17.	Discuss large scale commercial production of penicillin G with the help of a suitable flow chart	CO3
18.	Draw and discuss schematic diagram for batch production of penicillin.	CO3
CO1:	Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability).	
CO2:	Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.	
CO3:	Exhibiting knowledge of the systems used for bioresources and bioresource technology.	
CO4:	Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.	